Summer 2026 Master's Applicant Interview Topics

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Depending on the position you are applying for, please prepare the corresponding topic below before the interview. We are mainly interested in your understanding, technical judgment, and ability to explain implementation details clearly.

Track 1: TurtleBot Applicants

Preparation task: Watch the following video carefully: https://youtu.be/nbqg_tmVgjU

What to prepare for the interview:

- Be ready to explain what the robot components are.
- Be ready to discuss your understanding of the robotics system shown in the video.
- Identify possible strengths, limitations, failure modes, and practical improvements.
- Be prepared for follow-up technical questions during the interview based on your interpretation of the video.

Small coding exercise: In addition to the video discussion, please implement a simple ROS 2 package with the following three nodes.

- A `mock sensor node` that periodically publishes fake sensor data, such as a distance value, to `/mock_range` using `std_msgs/Float32`.
- A `processor node` that subscribes to `/mock_range`, compares the value against a threshold, and publishes a status message to `/status` using `std_msgs/String`.
- The status should be something simple such as `"CLEAR"` and `"BLOCKED"`.
- A `logger node` that subscribes to `/status` and prints status changes.
- Provide a launch file so the whole package can be started with one command.
- Make `threshold` configurable through a ROS 2 parameter.
- Include a short `README` explaining how to build and run the package.

Track 2: Spatial Reasoning Applicants

Preparation task: Install AI Habitat and work with the fixed HM3D scene `hm3d/minival/00800-TEEsavR23oF`. Try to extract first-person data that imitates a human walking through the environment, and then build a basic perception pipeline on top of that data.

Suggested task scope:

- Install Habitat-Sim / Habitat-Lab and become familiar with the environment setup.
- Use the specific scenario `hm3d/minival/00800-TEEsavR23oF` and collect an egocentric first-person trajectory that approximates human walking in the scene.
- Save a sequence of RGB frames from that trajectory as your working data.
- Use YOLO-World to detect objects in the collected frames.
- Use NanoSAM to produce object segmentation masks.
- Use Depth Pro to estimate pixel-level depth for the scene and analyze object depth together with the segmentation results.
- Be ready to explain your pipeline design, prompts or object categories, qualitative results, and failure cases.

Useful references:

- AI Habitat overview: https://aihabitat.org/
- Habitat-Sim documentation: https://aihabitat.org/docs/habitat-sim/
- Habitat-Lab documentation: https://aihabitat.org/docs/habitat-lab/

- HM3D room page: https://aihabitat.org/datasets/hm3d/00800-TEEsavR23oF/index.html
- YOLO-World official code: https://github.com/AILab-CVC/YOLO-World
- YOLO-World paper: CVPR 2024 paper link
- NanoSAM official code: https://github.com/NVIDIA-AI-IOT/nanosam
- Depth Pro official code: https://github.com/apple/ml-depth-pro
- Depth Pro research page: https://machinelearning.apple.com/research/depth-pro

Interview Expectation

The goal is not necessarily to build a fully polished system. We care more about whether you can learn quickly, make reasonable technical choices, and explain your observations clearly.

For both tracks, please be prepared to explain what you completed for the assignment, how you approached it, what worked, what did not work, and what you would improve next. Please bring your notes, code, or visual results if you attempted the implementation task.